

HeartLang-Based Multi-Label ECG Classification with Signal Preprocessing and Full Fine-Tuning on PTB-XL

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Abstract

Electrocardiogram (ECG) classification is an important task for automated cardiac abnormality detection, but effective representation of complex ECG waveforms remains challenging. In this work, we investigate HeartLang, a pretrained ECG representation model, for multi-label classification on the PTB-XL Superdiagnostic dataset. The proposed framework applies ECG preprocessing consisting of a 0.5–40 Hz Butterworth band-pass filter followed by lead-wise z-score normalization, after which QRS-based representations are generated and provided to HeartLang for task-specific full fine-tuning. The classification task considers five diagnostic categories: NORM, MI, STTC, CD, and HYP. Experiments are conducted using a fixed predefined training/validation/test split and three random seeds, with model selection based on validation ROC-AUC. The proposed preprocessing and full fine-tuning configuration achieves a mean test ROC-AUC of 0.9126 ± 0.0010 and an F1-score of 0.6791 ± 0.0062 across the three seeds. Compared with HeartLang without the proposed preprocessing, the preprocessing pipeline improves ROC-AUC from 0.9020 ± 0.0013 to 0.9126 ± 0.0010 . The results demonstrate that preprocessing combined with pretrained ECG representations can provide effective and stable performance for multi-label ECG classification.

Keywords: HeartLang, electrocardiogram, ECG classification, PTB-XL, QRS tokenization, multi-label classification, deep learning

1 Introduction

Electrocardiography (ECG) is a widely used non-invasive technique for recording the electrical activity of the heart and plays an important role in the assessment of cardiovascular abnormalities. A standard clinical ECG typically consists of multiple leads that capture complementary views of cardiac electrical activity, providing information related to rhythm, conduction, ischemic changes,

and other pathological conditions. The availability of large-scale annotated ECG datasets has consequently enabled the development of computational methods for automated interpretation and computer-aided cardiovascular diagnosis [1].

The increasing volume and complexity of ECG recordings have motivated the use of machine learning and deep learning methods for automated ECG classification. Convolutional neural networks (CNNs), recurrent neural networks (RNNs), and

related architectures have demonstrated the ability to learn discriminative patterns directly from ECG waveforms and have reduced the need for manually designed features. However, ECG signals contain morphological and temporal information distributed across multiple cardiac cycles and leads, making the representation of long-range dependencies an important challenge for automated analysis. Recent reviews have highlighted the effectiveness of deep learning for ECG-based cardiovascular disease detection while also identifying challenges related to generalization, interpretability, data requirements, and robustness [2].

Transformer-based models have recently emerged as an alternative for ECG representation learning because self-attention mechanisms can model relationships between elements of a sequence over comparatively long temporal contexts. This development has further motivated the use of pretrained ECG representation models and foundation-model approaches, in which knowledge learned from large-scale ECG data can be adapted to downstream clinical classification tasks. Recent literature indicates a growing interest in transformer and large-language-model-inspired approaches for ECG analysis, while also emphasizing the importance of effective ECG representations, appropriate preprocessing, and reliable evaluation protocols [3].

In this study, we investigate the application of the pretrained HeartLang model to multi-label ECG classification on the PTB-XL Superdiagnostic task. Rather than modifying the underlying HeartLang architecture, the study focuses on adapting the pretrained model to the target task through task-specific fine-tuning and on investigating the effect of ECG signal preprocessing before QRS tokenization. The preprocessing pipeline consists of Butterworth band-pass filtering followed by lead-wise z-score normalization. The resulting ECG representations are used for downstream HeartLang-based classification of five Superdiagnostic categories: normal ECG (NORM), myocardial infarction (MI), ST/T changes (STTC), conduction disturbance (CD), and hypertrophy (HYP).

The study provides a systematic evaluation of the resulting framework through three complementary experimental directions. First, different

HeartLang adaptation strategies, including linear probing, adapter-based tuning, and full fine-tuning, are investigated. Second, the proposed preprocessing pipeline is evaluated through comparison with HeartLang without preprocessing and against conventional baseline architectures, including a linear classifier, multilayer perceptron (MLP), CNN, gated recurrent unit (GRU), and ResNet. Third, the robustness and practical behavior of the framework are examined through data-efficiency experiments, class-wise ROC-AUC analysis, confusion-matrix analysis, G-Mean evaluation, and repeated experiments using three random seeds. This evaluation is intended to determine whether appropriate signal preprocessing can improve the adaptation of pretrained ECG representations for multi-label diagnostic classification.

2 Literature Review

2.1 Automated ECG Classification

Electrocardiogram (ECG) interpretation has traditionally relied on expert assessment of waveform morphology, rhythm characteristics, and clinically relevant intervals. Although expert interpretation remains an important component of cardiovascular diagnosis, the increasing volume of ECG recordings has motivated the development of automated computer-aided analysis methods. The availability of large annotated ECG datasets has further enabled the development and systematic evaluation of machine learning approaches for automated ECG interpretation. In particular, the PTB-XL dataset established a large publicly accessible benchmark for 12-lead ECG analysis and provided structured multi-label diagnostic annotations, facilitating reproducible research in automated ECG classification [1].

Early automated approaches commonly relied on manually designed features derived from ECG morphology, rhythm, and temporal characteristics. These features were subsequently provided to conventional machine learning classifiers for diagnostic prediction. Such pipelines can perform effectively for specific tasks, but their reliance on handcrafted features may limit their ability to capture complex and distributed patterns present in multi-lead ECG signals. The development of deep learning methods has therefore shifted the

focus toward end-to-end approaches that learn task-relevant representations directly from ECG waveforms.

Large-scale studies have demonstrated the potential of deep neural networks for automated ECG interpretation. Hannun et al. showed that a deep neural network trained on ambulatory ECG recordings could achieve cardiologist-level performance for several rhythm classification tasks, demonstrating the feasibility of learning clinically relevant ECG representations directly from raw signals [4]. Subsequent research has extended deep learning approaches to a broad range of ECG applications, including arrhythmia detection, cardiovascular disease diagnosis, myocardial infarction detection, and other clinical prediction tasks. Systematic reviews have reported the increasing use of architectures such as convolutional neural networks, recurrent neural networks, gated recurrent units, and residual networks for ECG analysis [5].

Despite these advances, automated ECG classification remains challenging because ECG recordings contain temporal dependencies, morphological variations, multiple simultaneous diagnostic labels, and differences across patients and recording conditions. In addition, the performance of data-intensive deep learning approaches can depend strongly on the size and quality of the available labeled dataset. These challenges have motivated research into pretrained ECG representations and transformer-based approaches, which aim to learn more generalizable representations that can be adapted to downstream clinical tasks.

2.2 Deep Learning for ECG Analysis

The application of deep learning has substantially advanced automated ECG analysis by enabling models to learn discriminative representations directly from ECG signals rather than relying exclusively on handcrafted features. Convolutional neural networks (CNNs) have been widely adopted for ECG analysis because their local receptive fields and hierarchical feature extraction capabilities can capture characteristic waveform patterns from one-dimensional ECG signals. Recurrent neural networks (RNNs), including long short-term memory (LSTM) and gated recurrent

unit (GRU) architectures, have also been investigated to model temporal dependencies within ECG sequences [5].

Deep learning approaches have been applied to a wide range of ECG-related tasks, including arrhythmia detection, cardiovascular disease classification, myocardial infarction detection, and ECG phenotype prediction. CNN-based models can learn morphological characteristics such as waveform shape and local temporal patterns, whereas recurrent architectures are capable of incorporating sequential context across ECG observations. Hybrid architectures combining convolutional and recurrent components have also been explored to jointly exploit morphological and temporal information. Systematic reviews of ECG deep learning research indicate that CNNs, RNNs, GRUs, residual networks, and related architectures remain widely used across different ECG analysis applications [5].

The effectiveness of deep learning for ECG interpretation has also been demonstrated in large-scale clinical studies. Hannun et al. reported that a deep neural network trained on ambulatory ECG recordings achieved cardiologist-level performance for multiple rhythm classification tasks, highlighting the ability of deep neural networks to learn clinically meaningful ECG representations directly from signal data [4]. These developments demonstrate the potential of end-to-end representation learning for reducing dependence on manually engineered ECG features.

Despite their effectiveness, conventional deep learning models generally require substantial amounts of task-specific labeled data and computational resources for training. Their performance can also be sensitive to signal quality, preprocessing choices, dataset characteristics, and differences between training and deployment populations. Furthermore, learning general-purpose ECG representations separately for every downstream task can be computationally inefficient. These limitations have motivated the development of transfer-learning, transformer-based, and pretrained ECG representation approaches that can leverage knowledge learned from large-scale datasets and subsequently adapt it to specific clinical tasks.

2.3 Transformer-Based ECG Representation Learning

The limitations of conventional convolutional and recurrent architectures have motivated the application of Transformer-based models to ECG analysis. Transformers employ self-attention mechanisms to model relationships between different elements of a sequence, allowing information from distant temporal regions to contribute to the learned representation. This property is particularly relevant to ECG signals, in which diagnostic information may be distributed across multiple cardiac cycles and across different leads.

Early studies investigated Transformer architectures directly for automated ECG classification. These approaches demonstrated that self-attention can be used to capture temporal relationships within ECG recordings and can provide competitive performance for heartbeat and rhythm classification tasks [6]. Subsequent work has explored Transformer architectures for multi-lead ECG classification, including approaches that combine local waveform information with global temporal dependencies [7].

Beyond supervised classification, recent research has increasingly focused on self-supervised representation learning for ECG signals. Instead of directly optimizing a model for a single labeled task, self-supervised approaches learn general representations from large collections of unlabeled or weakly annotated ECG recordings and subsequently adapt these representations to downstream applications. Masked modeling and other pretraining strategies have been investigated for learning representations that capture meaningful ECG structure while reducing reliance on large quantities of manually annotated data [8].

This shift toward pretrained representations is particularly important for clinical ECG applications because obtaining high-quality diagnostic annotations can require substantial effort from trained medical professionals. Recent work has consequently explored ECG foundation models that use large-scale pretraining to learn generalizable signal representations before task-specific adaptation. Such approaches provide a potentially more efficient alternative to training conventional deep learning models independently for each downstream ECG classification task.

2.4 ECG Foundation Models and HeartLang

The development of ECG foundation models represents a shift from task-specific deep learning toward pretrained representations that can be adapted to multiple downstream ECG analysis tasks. Rather than learning all task-relevant features from a limited labeled dataset, foundation models can first learn general ECG representations from large-scale data and subsequently be fine-tuned for specific diagnostic objectives. This paradigm is particularly attractive for ECG analysis because obtaining high-quality diagnostic annotations requires substantial clinical expertise and effort.

Several recent studies have demonstrated the potential of foundation models for ECG analysis. Transformer-based foundation-model approaches have been investigated for arrhythmia detection and classification, demonstrating that heartbeat-level representations can be adapted to downstream tasks using relatively limited labeled data [9]. Other large-scale ECG foundation models have been developed using millions of ECG recordings and have demonstrated broad diagnostic capabilities across multiple conditions and datasets [10]. These studies indicate that large-scale pretraining can provide transferable representations and improve label efficiency for downstream ECG applications.

A related line of research investigates language-inspired representations of ECG signals. HeartLang introduces an ECG language-processing framework that treats individual heartbeats as words and sequences of cardiac rhythms as sentences. Its QRS-Tokenizer converts ECG waveforms into semantically meaningful heartbeat-based token sequences, after which a Transformer-based model learns representations at both waveform-form and rhythm levels [11]. This formulation differs from conventional approaches that represent ECG recordings using fixed-size temporal windows, as it explicitly incorporates cardiac-cycle structure into the representation process.

Despite the progress of ECG foundation models, downstream performance can still depend on the quality and consistency of the ECG signals provided to the pretrained model. Most foundation-model studies primarily focus on

model architecture, pretraining scale, or downstream adaptation, while the effect of task-specific signal preprocessing before tokenization and fine-tuning remains less systematically investigated. In addition, comparative studies that jointly examine fine-tuning strategies, data efficiency, and class-wise diagnostic performance can provide further insight into the practical adaptation of pretrained ECG representations.

Motivated by these observations, the present study investigates the adaptation of HeartLang to multi-label PTB-XL Superdiagnostic ECG classification. The underlying HeartLang architecture is retained, while an ECG preprocessing pipeline is introduced before QRS tokenization. The study further evaluates different fine-tuning strategies, the effect of training data availability, and class-wise diagnostic performance to provide a comprehensive assessment of preprocessing-enhanced HeartLang adaptation.

2.5 HeartLang

HeartLang is a pretrained self-supervised ECG representation learning framework introduced to model ECG signals using a language-inspired representation. The framework interprets individual heartbeats as ‘words’ and sequences of cardiac rhythms as ‘sentences’, thereby incorporating both local waveform morphology and higher-level temporal context into the learned representation [11].

A central component of HeartLang is the QRS-Tokenizer, which transforms ECG waveforms into heartbeat-based token sequences. This representation is designed to preserve the structural organization of cardiac cycles while providing a sequence format that can be processed by a Transformer-based encoder. HeartLang subsequently learns generalized ECG representations from these tokenized sequences through self-supervised pretraining, enabling the resulting model to be adapted to downstream ECG analysis tasks [11].

The pretrained representation provided by HeartLang offers an alternative to training task-specific ECG models entirely from scratch. By initializing a downstream model with pretrained ECG representations, task-specific supervised learning can focus on adapting the learned representations to the target diagnostic problem. The official HeartLang implementation also provides

pretrained checkpoints and supports downstream fine-tuning on multiple ECG datasets, including PTB-XL [11].

In the present study, HeartLang is used as the pretrained backbone for multi-label classification on the PTB-XL Superdiagnostic task. The underlying HeartLang architecture is retained without architectural modification. Instead, the study investigates task-specific adaptation through full fine-tuning and evaluates the effect of ECG signal preprocessing before QRS tokenization. This design allows the contribution of signal preprocessing and fine-tuning strategy to be evaluated independently from the pretrained HeartLang architecture.

2.6 Research Gap

The existing literature demonstrates substantial progress in automated ECG classification, deep learning-based ECG analysis, Transformer-based representation learning, and ECG foundation models. However, several aspects of adapting pretrained ECG representations to downstream diagnostic tasks remain insufficiently explored. In particular, the effect of explicit ECG signal preprocessing on the performance of pretrained ECG language models has received comparatively limited systematic evaluation.

Although pretrained models such as HeartLang provide general-purpose ECG representations, their downstream performance can depend on the quality and characteristics of the ECG signals provided to the tokenization stage. Furthermore, the relative effectiveness of different fine-tuning strategies, the amount of labeled training data required for effective adaptation, and the resulting performance across individual diagnostic classes are important practical considerations for downstream ECG classification.

To address these gaps, the present study systematically evaluates HeartLang on the PTB-XL Superdiagnostic multi-label classification task. The study investigates the effect of ECG preprocessing before QRS-based tokenization and compares downstream adaptation strategies using a pretrained HeartLang backbone. In addition, the study evaluates data efficiency using different fractions of the training set and compares HeartLang with conventional deep learning baselines. Performance is assessed using overall and per-class

metrics to provide a comprehensive evaluation of the resulting ECG representations for multi-label diagnosis.

3 Proposed Methodology

3.1 Overall Framework

The proposed framework employs HeartLang for multi-label electrocardiogram (ECG) classification on the PTB-XL Superdiagnostic dataset. The overall methodology consists of ECG preprocessing, QRS-based tokenization, representation learning using the pretrained HeartLang model, and task-specific fine-tuning for multi-label classification. In the proposed configuration, the input ECG recordings are first processed to reduce noise and baseline variations. The resulting signals are subsequently transformed into QRS-based representations that are provided to the pretrained HeartLang model. The model is then fully fine-tuned on the target classification task, with the five Superdiagnostic classes, namely NORM, MI, STTC, CD, and HYP, considered as the prediction targets. The complete experimental protocol uses fixed predefined training, validation, and held-out test sets, while data-efficiency experiments are performed by randomly subsampling only the training set.

3.2 ECG Preprocessing

To improve the quality and consistency of the ECG signals provided to the HeartLang framework, a preprocessing pipeline was applied prior to QRS tokenization. The preprocessing consisted of two primary operations: band-pass filtering and lead-wise normalization. First, a Butterworth band-pass filter with cutoff frequencies of 0.5–40 Hz was applied to attenuate baseline wander and high-frequency noise while preserving the clinically relevant frequency components of the ECG signal. Subsequently, lead-wise z-score normalization was performed using statistics computed from the training data. This normalization reduces differences in signal scale across ECG leads while preventing information from the validation and test sets from influencing the preprocessing parameters.

The preprocessed ECG signals were then transformed into QRS-based token representations and provided as input to the pretrained HeartLang model. The same preprocessing procedure was applied consistently to the corresponding training, validation, and held-out test data. The effect of this preprocessing pipeline was evaluated by comparing HeartLang performance with and without preprocessing under the same full fine-tuning configuration.

3.3 QRS Tokenization and HeartLang Representation

Following preprocessing, the ECG signals were transformed into QRS-based representations compatible with the pretrained HeartLang framework. The QRS tokenization process converts the temporal ECG signal into a structured representation that can be processed by the pretrained model. This representation enables the model to capture informative patterns from the ECG waveform while providing a consistent input format for downstream classification.

The resulting QRS-based representations were used as inputs to the pretrained HeartLang model initialized from the available pretrained checkpoint. Rather than training the representation model from scratch, the proposed framework leverages the pretrained HeartLang representations and adapts them to the PTB-XL Superdiagnostic classification task through supervised fine-tuning. The same representation procedure was maintained across the training, validation, and held-out test sets to ensure consistency during model development and evaluation.

3.4 Hamilton QRS Detection

Following ECG preprocessing, the Hamilton QRS detection algorithm was employed to identify the QRS complexes from the filtered ECG recordings. The Hamilton detector was selected to provide robust R-peak detection in the presence of residual noise and morphological variations in the ECG signals. The detection procedure exploits amplitude and slope information together with temporal constraints to distinguish QRS complexes from other ECG waveform components.

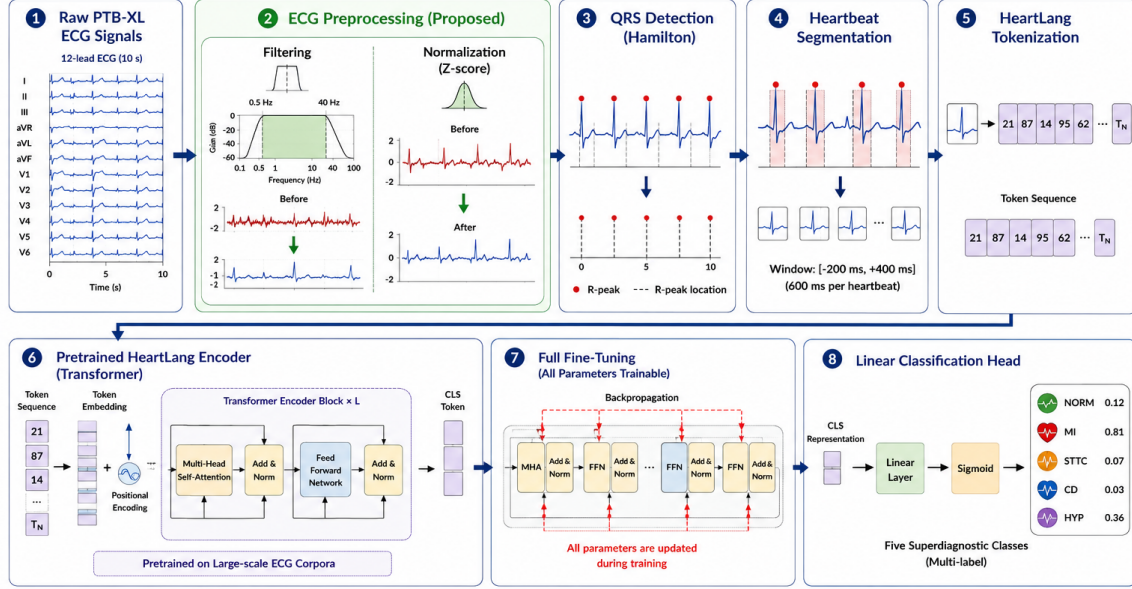


Fig. 1 Overview of the proposed HeartLang-based ECG classification methodology.

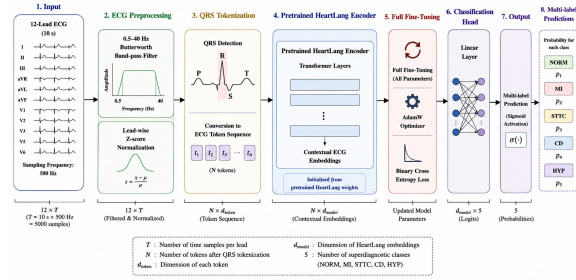


Fig. 2 Architecture of the HeartLang-based ECG classification framework.

For each preprocessed ECG recording, the Hamilton detector was applied to identify candidate R-peaks and determine the temporal locations of the QRS complexes. Adaptive detection thresholds were used to accommodate variations in signal amplitude across recordings. The detected R-peaks served as reference points for extracting the corresponding QRS regions from the preprocessed ECG signals.

The detected QRS complexes were subsequently segmented into fixed-length signal regions and used as the basis for the downstream QRS tokenization process. This procedure converts the

continuous ECG recordings into structured QRS-centered representations while preserving the morphological information relevant to multi-label cardiac diagnosis. The resulting QRS-based representations were then provided to the pretrained HeartLang model for representation learning and task-specific fine-tuning.

3.5 HeartLang Fine-Tuning and Multi-Label Classification

The QRS-based ECG representations obtained from the preceding processing stages were provided to the pretrained HeartLang model for task-specific fine-tuning. A pretrained HeartLang checkpoint was used as the initialization of the model, thereby allowing the downstream classification task to benefit from the representations learned during pretraining.

For the final experiments, full fine-tuning was employed, in which all trainable parameters of the HeartLang model were optimized for the target classification task. A task-specific linear classification head was used to map the learned ECG representation to the five PTB-XL Superdiagnostic diagnostic categories: NORM, MI, STTC, CD, and HYP.

The classification task was formulated as a multi-label problem because a single ECG recording may be associated with more than one diagnostic category. Accordingly, the classification head produces an independent prediction for each of the five diagnostic classes. The model therefore generates a five-dimensional output corresponding to the presence or absence of each diagnostic category.

Model training was performed using the pre-defined training and validation splits, while the held-out test set was kept unchanged for final evaluation. The best model checkpoint was selected according to validation ROC-AUC during training. To assess the stability of the reported results, the final experiments were conducted using three random seeds (0, 1, and 2), and the test performance was reported as mean $\pm$ standard deviation across the three runs.

The proposed preprocessing configuration was applied before QRS tokenization in the final HeartLang experiments. Specifically, the ECG signals were processed using a 0.5–40 Hz Butterworth band-pass filter followed by lead-wise z-score normalization using training-set statistics. The resulting QRS-based representations were subsequently passed to HeartLang for full fine-tuning and multi-label classification.

3.6 Dataset Partitioning and Experimental Protocol

A fixed, pre-defined train/validation/test partition of the PTB-XL Superdiagnostic dataset was used throughout the experiments. The training split contained 17,084 ECG samples, while the validation and held-out test splits contained 2,146 and 2,158 samples, respectively. The HeartLang fine-tuning procedure loaded these stage-specific datasets directly, and the train, validation, and test partitions were not regenerated during model training.

For the data-efficiency experiments, only the training partition was randomly subsampled while the validation and test sets remained unchanged. Training fractions of 10%, 25%, 50%, and 100% corresponded to 1,708, 4,271, 8,542, and 17,084 training samples, respectively. This protocol enabled the effect of training-data availability on HeartLang performance to be evaluated under a consistent validation and held-out test setting.

The final HeartLang experiments were repeated using three random seeds, namely 0, 1, and 2. The best checkpoint for each run was selected according to validation ROC-AUC, after which the selected model was evaluated on the fixed held-out test set. The resulting performance metrics were summarized as mean $\pm$ standard deviation across the three seeds.

3.7 Training Configuration

The HeartLang model was initialized using pre-trained weights and subsequently fine-tuned on the PTB-XL Superdiagnostic classification task. Full fine-tuning was employed for the final experiments, allowing the model parameters and the task-specific classification head to be optimized jointly for the target multi-label classification problem. Training was performed using the pre-defined training and validation partitions, with the held-out test set reserved exclusively for final evaluation.

The final experiments were conducted using three random seeds (0, 1, and 2) to assess the stability of model performance. During training, model checkpoints were evaluated on the validation set, and the checkpoint achieving the highest validation ROC-AUC was selected for subsequent test-set evaluation. The same experimental protocol was maintained across the three seeds.

The selected training configurations are summarized in Table 1.

The principal training and evaluation settings used for the HeartLang full fine-tuning experiments are summarized in Table 1.

3.8 Data Split and Evaluation Protocol

A fixed predefined split of the PTB-XL Superdiagnostic dataset was used for all HeartLang experiments. The training, validation, and held-out test sets contained 17,084, 2,146, and 2,158 ECG samples, respectively. The training and validation sets were used for model optimization and checkpoint selection, while the test set was kept completely held out for final performance evaluation.

For the data-efficiency experiments, only the training split was randomly subsampled to obtain different fractions of the available training data,

Table 1 Training Configuration for HeartLang Full Fine-Tuning

Parameter	Configuration
Pretrained model	HeartLang
Pretrained checkpoint	checkpoint-200.pth
Fine-tuning strategy	Full fine-tuning
Dataset	PTB-XL Superdiagnostic
Training samples	17,084
Validation samples	2,146
Test samples	2,158
Random seeds	0, 1, 2
Checkpoint selection	Best validation ROC-AUC
Preprocessing	0.5–40 Hz Butterworth band-pass filtering + lead-wise z-score normalization
Optimizer	AdamW
Learning rate	5×10^{-3}
Batch size	1
Epochs	10
Weight decay	0.05
Loss function	Binary cross-entropy
Decision threshold	0.5

namely 10%, 25%, 50%, and 100%. The validation and test sets remained unchanged across all data fractions. This protocol ensured that performance differences across training-data fractions were attributable to the amount of training data rather than changes in the evaluation sets.

The main HeartLang experiments were conducted using three random seeds (0, 1, and 2). For each seed, the best model checkpoint was selected according to validation ROC-AUC and subsequently evaluated on the fixed held-out test set. Final performance is reported as mean $\pm$ standard deviation across the three seeds. The primary evaluation metric was ROC-AUC, with accuracy, precision, recall, F1-score, and test loss reported as complementary performance measures.

4 Results and Evaluation

4.1 Evaluation Metrics

The performance of the proposed HeartLang-based ECG classification framework was evaluated using multiple complementary metrics to account for the multi-label nature and class imbalance of the PTB-XL Superdiagnostic dataset. Macro Receiver Operating Characteristic Area Under the Curve (Macro ROC-AUC) was used as the primary evaluation metric because it gives equal

importance to the individual diagnostic categories. In addition, Accuracy, Precision, Recall, and F1-score were considered to provide a broader assessment of classification performance. Per-class ROC-AUC values were further analyzed to examine the discriminative capability of the model for individual diagnostic categories.

For a binary diagnostic category, the Receiver Operating Characteristic (ROC) curve describes the relationship between the true positive rate and the false positive rate across different classification thresholds. The Area Under the ROC Curve (ROC-AUC) summarizes this discriminative ability, with larger values indicating better separation between positive and negative samples. For the five Superdiagnostic categories, the macro ROC-AUC is obtained by averaging the ROC-AUC values calculated independently for each category:

$$\text{Macro ROC-AUC} = \frac{1}{C} \sum_{c=1}^C \text{AUC}_c, \quad (1)$$

where $C = 5$ represents the number of Superdiagnostic categories and AUC_c denotes the ROC-AUC of category c .

Precision and Recall were used to characterize the correctness and sensitivity of positive predictions, respectively. They are defined in terms of true positives (TP), false positives (FP), and false negatives (FN) as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN}. \quad (3)$$

The F1-score combines Precision and Recall through their harmonic mean and is therefore useful when both types of classification error need to be considered:

$$F1 = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (4)$$

Accuracy measures the proportion of predictions that are correctly classified under the selected decision threshold. In this study, a threshold of 0.5 was used to convert the independently predicted class probabilities into binary labels. The confusion matrix was then used to examine class-wise prediction outcomes through true

positives, false positives, true negatives, and false negatives.

For additional class-wise analysis, sensitivity and specificity were used to derive the geometric mean (G-Mean), which provides a balanced measure of positive and negative classification performance:

$$\text{G-Mean} = \sqrt{\text{Sensitivity} \times \text{Specificity}}. \quad (5)$$

4.2 Overall Performance

The overall performance of the proposed HeartLang-based framework was evaluated on the held-out PTB-XL Superdiagnostic test set using the complete training set and three independent random seeds. The model incorporating the proposed ECG preprocessing pipeline and Full Fine-Tuning achieved a Macro ROC-AUC of 0.9126 ± 0.0010 . The corresponding Accuracy, Precision, Recall, and F1-score were 0.5779 ± 0.0093 , 0.7608 ± 0.0121 , 0.6549 ± 0.0099 , and 0.6791 ± 0.0062 , respectively, while the reported test loss was 0.3006 ± 0.0040 . The relatively small standard deviations across the three runs indicate stable performance across random seeds.

To quantify the contribution of ECG preprocessing, the proposed model was also evaluated without the preprocessing stage while retaining the same HeartLang Full Fine-Tuning strategy. Without preprocessing, the model achieved a Macro ROC-AUC of 0.9020 ± 0.0013 , with Accuracy, Precision, Recall, and F1-score of 0.5618 ± 0.0135 , 0.7358 ± 0.0172 , 0.6662 ± 0.0107 , and 0.6792 ± 0.0051 , respectively. With preprocessing, the Macro ROC-AUC increased to 0.9126 ± 0.0010 , corresponding to an absolute improvement of 0.0106 in ROC-AUC.

These results demonstrate that the pretrained HeartLang representation can achieve strong downstream performance when fully adapted to the PTB-XL Superdiagnostic classification task. More importantly, the improvement observed after applying ECG preprocessing indicates that appropriate signal conditioning before QRS tokenization can further improve the quality of the representations used for downstream classification. The remaining sections provide detailed comparisons with baseline models, fine-tuning

strategies, data-efficiency settings, and class-wise diagnostic performance.

All major experiments were conducted using three random seeds, and the reported performance values are presented as mean $\pm$ standard deviation to characterize experimental variability and stability.

Figure 3 shows the training loss and validation ROC-AUC across training epochs. For the HeartLang experiments, the best checkpoint was selected according to validation ROC-AUC.

4.3 Baseline Model Comparison

To assess the effectiveness of the proposed HeartLang-based framework, its performance was compared with five conventional deep learning baselines: Linear Classifier, Multi-Layer Perceptron (MLP), Convolutional Neural Network (CNN), Gated Recurrent Unit (GRU), and Residual Network (ResNet). In addition, two HeartLang configurations were evaluated, namely HeartLang with Full Fine-Tuning without preprocessing and HeartLang with the proposed ECG preprocessing pipeline followed by Full Fine-Tuning. All models were evaluated using the same held-out PTB-XL Superdiagnostic test set and the same multi-label evaluation protocol.

Among the conventional baseline models, the Linear Classifier obtained the lowest Macro ROC-AUC of 0.7360 ± 0.0036 , while the MLP achieved 0.8000 ± 0.0028 . The CNN and GRU models obtained Macro ROC-AUC values of 0.8749 ± 0.0016 and 0.8827 ± 0.0029 , respectively. ResNet achieved the strongest performance among the conventional baselines, with a Macro ROC-AUC of 0.8868 ± 0.0009 .

HeartLang without preprocessing achieved a substantially higher Macro ROC-AUC of 0.9020 ± 0.0013 , demonstrating the effectiveness of the pretrained ECG representation compared with the conventional architectures. After introducing the proposed ECG preprocessing pipeline, the Macro ROC-AUC increased further to 0.9126 ± 0.0010 . This represents an absolute improvement of 0.0106 in Macro ROC-AUC over the corresponding HeartLang configuration without preprocessing.

Figure 4 compares the test ROC-AUC of the seven evaluated models, with values reported as

Training Dynamics Comparison Across Baseline Models and the Proposed HeartLang Framework

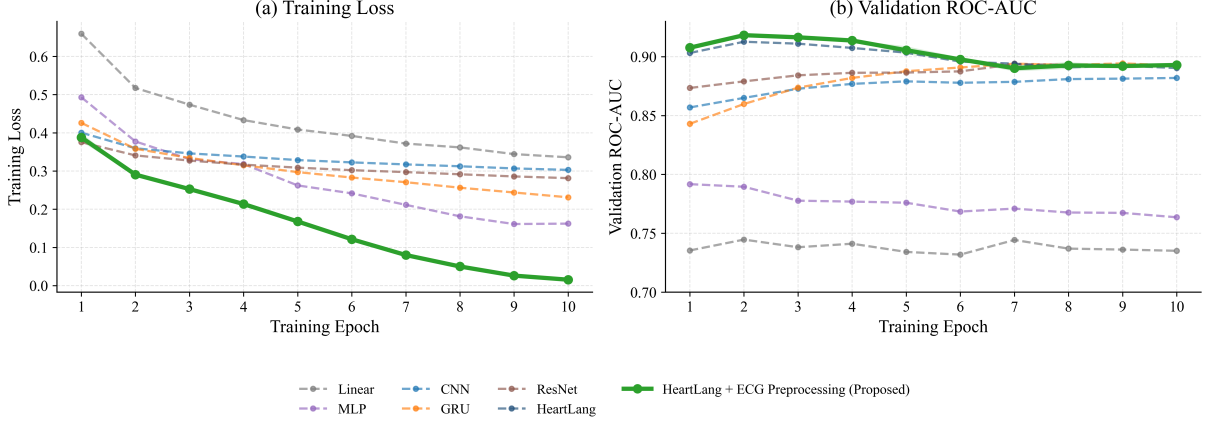


Fig. 3 Training dynamics of the evaluated models showing training loss and validation ROC-AUC across training epochs.

mean $\pm$ standard deviation across three random seeds.

The proposed preprocessed HeartLang model also achieved the highest Accuracy and Precision among the evaluated models, with values of 0.5779 ± 0.0093 and 0.7608 ± 0.0121 , respectively. Its Recall was 0.6549 ± 0.0099 and its F1-score was 0.6791 ± 0.0062 . The unprocessed HeartLang model achieved a marginally higher F1-score of 0.6792 ± 0.0051 , indicating that the preprocessing improvement is primarily reflected in discriminative performance rather than a uniform increase across every threshold-dependent metric.

Compared with the strongest conventional baseline, ResNet, the proposed HeartLang framework improves Macro ROC-AUC from 0.8868 ± 0.0009 to 0.9126 ± 0.0010 , corresponding to an absolute gain of approximately 2.58 percentage points. Overall, the comparison demonstrates that pretrained ECG-specific representations provide a strong basis for downstream classification, while the proposed preprocessing stage further improves the discriminative capability of the HeartLang-based framework.

4.4 Fine-Tuning Ablation Study

To investigate the effect of different adaptation strategies on the pretrained HeartLang model, an

ablation study was conducted using Linear Probing, Adapter-based Fine-Tuning, and Full Fine-Tuning. The three strategies differ in the extent to which the pretrained HeartLang encoder is updated during downstream supervised learning.

In the Linear Probing setting, the pretrained HeartLang encoder was kept frozen and only the linear classification head was trained. This approach achieved a Macro ROC-AUC of 0.8719 ± 0.0004 and an F1-score of 0.5845 ± 0.0008 . The result demonstrates that the pretrained HeartLang representations already provide useful information for the PTB-XL Superdiagnostic task, even when the encoder parameters are not updated.

Adapter-based Fine-Tuning was also investigated as a lightweight adaptation strategy. In this experiment, trainable adapter components were introduced while the majority of the pretrained encoder remained frozen. The initial experiment using seed 0 achieved a Macro ROC-AUC of 0.8408 and an F1-score of 0.4859. Since this performance was substantially lower than both Linear Probing and Full Fine-Tuning, the adapter experiment was not continued for additional random seeds.

Full Fine-Tuning, in which the pretrained HeartLang encoder and classification head were jointly optimized, achieved the strongest performance in the ablation study, with a Macro ROC-AUC of 0.9020 ± 0.0013 and an F1-score of

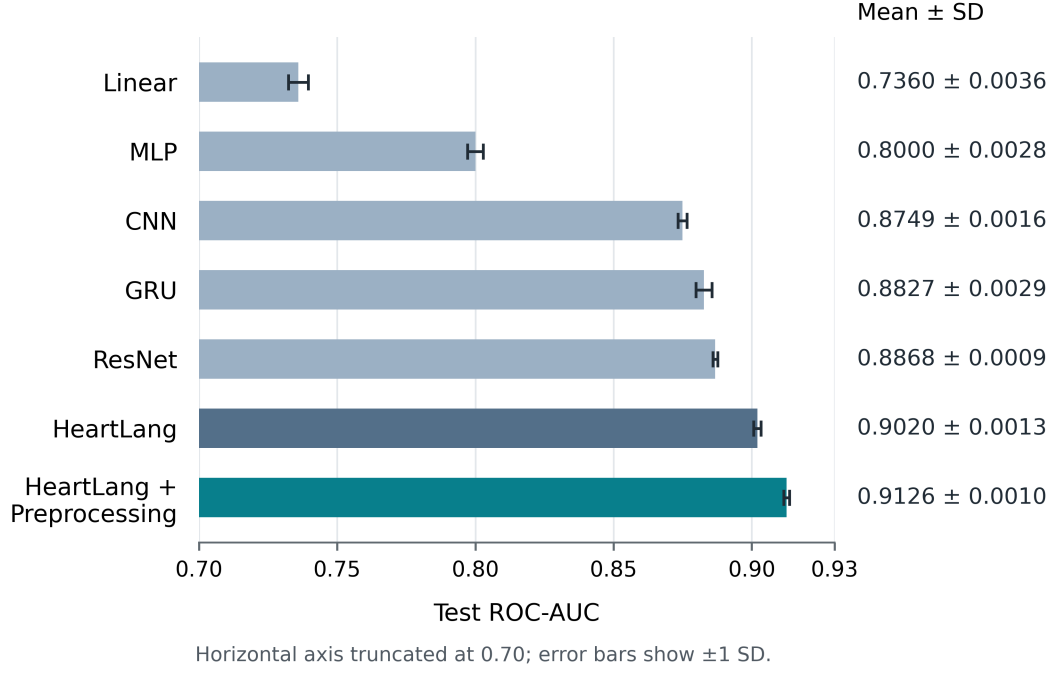


Fig. 4 Comparison of test ROC-AUC across baseline models and the proposed HeartLang framework with ECG preprocessing. Values represent mean $\pm$ standard deviation across three random seeds.

Table 2 Performance comparison of baseline and HeartLang models on the PTB-XL Superdiagnostic test set.

Model	ROC-AUC	Accuracy	Precision	Recall	F1-score
Linear	0.7360 ± 0.0036	0.3458 ± 0.0003	0.5140 ± 0.0226	0.4865 ± 0.0256	0.4967 ± 0.0039
MLP	0.8000 ± 0.0028	0.4384 ± 0.0146	0.6562 ± 0.0093	0.4740 ± 0.0218	0.5391 ± 0.0168
CNN	0.8749 ± 0.0016	0.4838 ± 0.0626	0.7297 ± 0.0208	0.5921 ± 0.0077	0.6407 ± 0.0141
GRU	0.8827 ± 0.0029	0.5545 ± 0.0049	0.7450 ± 0.0301	0.6000 ± 0.0163	0.6516 ± 0.0061
ResNet	0.8868 ± 0.0009	0.5479 ± 0.0100	0.7584 ± 0.0223	0.5845 ± 0.0240	0.6439 ± 0.0150
HeartLang (No Preprocessing)	0.9020 ± 0.0013	0.5618 ± 0.0135	0.7358 ± 0.0172	0.6662 ± 0.0107	0.6792 ± 0.0051
HeartLang + Preprocessing	0.9126 ± 0.0010	0.5779 ± 0.0093	0.7608 ± 0.0121	0.6549 ± 0.0099	0.6791 ± 0.0062

0.6792 $\pm$ 0.0051. Compared with Linear Probing, Full Fine-Tuning improved the Macro ROC-AUC by approximately 0.0301 and the F1-score by approximately 0.0947.

Overall, the ablation results indicate that updating the pretrained HeartLang encoder provides a substantial benefit for downstream multi-label ECG classification. Therefore, Full Fine-Tuning was selected as the primary adaptation strategy for the subsequent baseline comparison,

data-efficiency analysis, per-class evaluation, and preprocessing study.

4.5 Data-Efficiency Analysis

To investigate the data efficiency of the proposed HeartLang framework, the model was evaluated using different fractions of the available PTB-XL Superdiagnostic training data. Four training-data configurations were considered, corresponding to 10%, 25%, 50%, and 100% of the training set. The validation and held-out test sets were kept unchanged across all configurations to ensure a consistent evaluation protocol. The same ECG preprocessing pipeline and Full Fine-Tuning strategy were used for each training-data fraction.

The results show a consistent improvement in Macro ROC-AUC as the amount of training data increases. Using only 10% of the training data resulted in a Macro ROC-AUC of 0.8682. Increasing the training fraction to 25% and 50% improved the Macro ROC-AUC to 0.8773 and 0.8952, respectively. When the complete training set was used, the Macro ROC-AUC reached 0.9126.

Table 3 Data-efficiency performance of the proposed HeartLang framework using different fractions of the PTB-XL Superdiagnostic training set.

Training Data	Training Examples	Macro ROC-AUC
10%	1,708	0.8682
25%	4,271	0.8773
50%	8,542	0.8952
100%	17,084	0.9126

Fig. 5 illustrates this trend.

The improvement from 10% to 25% training data was 0.0091, while the increase from 25% to 50% was 0.0179. A further increase from 50% to 100% produced an improvement of 0.0174. These results indicate that additional labeled ECG data continues to improve downstream performance, with the complete training set providing the highest observed Macro ROC-AUC.

Overall, the data-efficiency experiment demonstrates that the pretrained HeartLang representation remains effective even when only a fraction of the available labeled training data is used. However, the observed performance also indicates

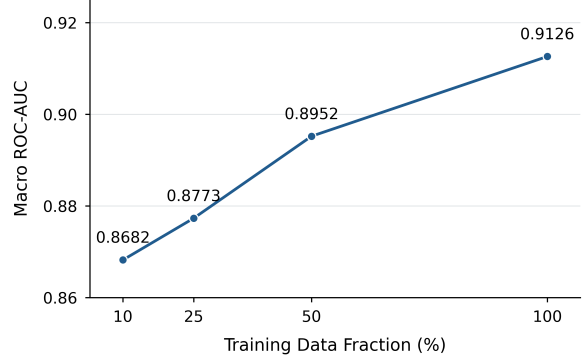


Fig. 5 Effect of training data fraction on the Macro ROC-AUC of the proposed HeartLang framework.

that additional labeled data remains beneficial for further improving the downstream classification performance.

4.6 Per-Class Performance

To examine the diagnostic performance of the proposed framework across individual categories, a class-wise evaluation was performed on the five PTB-XL Superdiagnostic labels: Normal (NORM), Myocardial Infarction (MI), ST/T Change (STTC), Conduction Disturbance (CD), and Hypertrophy (HYP). Since the task is multi-label classification, the ROC-AUC for each category was calculated independently using a one-versus-rest evaluation strategy. The class-wise results provide a more detailed assessment than the macro-average ROC-AUC alone.

The proposed HeartLang framework with ECG preprocessing achieved strong performance across all five diagnostic categories. The highest ROC-AUC was obtained for Conduction Disturbance (CD), with a value of 0.9369 ± 0.0009 , followed by Hypertrophy (HYP) with 0.9255 ± 0.0009 . ST/T Change (STTC) and Normal (NORM) achieved ROC-AUC values of 0.9091 ± 0.0025 and 0.9021 ± 0.0028 , respectively. Myocardial Infarction (MI) obtained the lowest class-wise ROC-AUC of 0.8908 ± 0.0040 , while still maintaining a ROC-AUC above 0.89.

The relatively small standard deviations across the three random seeds indicate stable class-wise performance. The macro-average ROC-AUC of 0.9126 ± 0.0010 reflects the consistently strong discriminative performance across the complete

Table 4 Per-class ROC-AUC of the proposed HeartLang framework with ECG preprocessing on the PTB-XL Superdiagnostic test set.

Diagnostic Class	ROC-AUC
NORM	0.9021 ± 0.0028
MI	0.8908 ± 0.0040
STTC	0.9091 ± 0.0025
CD	0.9369 ± 0.0009
HYP	0.9255 ± 0.0009
Macro Average	0.9126 ± 0.0010

set of diagnostic categories. The higher performance observed for CD and HYP suggests that the learned representations capture discriminative patterns for these categories particularly effectively, whereas MI presents comparatively greater classification difficulty.

Figure 6 presents the ROC curves for all five Superdiagnostic categories. The class-wise ROC curves complement the aggregate evaluation by illustrating the trade-off between true-positive and false-positive rates across different classification thresholds.

4.7 Confusion Matrix and G-Mean Analysis

To further investigate the class-wise behavior of the proposed HeartLang framework, confusion matrix analysis was performed using the final preprocessed Full Fine-Tuning model. The analysis provides the numbers of true positive (TP), false positive (FP), true negative (TN), and false negative (FN) predictions for each of the five Superdiagnostic categories. Because the task is multi-label, each diagnostic category is considered independently when constructing the corresponding binary confusion matrix.

Figure 7 presents the confusion matrices for NORM, MI, STTC, CD, and HYP obtained from the representative seed-2 test run. Across the five categories, the model correctly identifies a large number of negative samples, resulting in relatively high specificity. The model also identifies a substantial proportion of positive samples, although the number of false negatives varies across diagnostic categories.

Among the five categories, Conduction Disturbance (CD) has the largest number of true positive predictions, while Myocardial Infarction

(MI) shows comparatively fewer true positive predictions and a larger proportion of false negatives. The NORM category contains a large number of correctly classified negative samples, reflecting the relatively large number of normal recordings in the test set.

To provide a balanced class-wise measure of positive and negative classification performance, the geometric mean (G-Mean) of sensitivity and specificity was also calculated. The resulting values are summarized in Table 5. Higher G-Mean values indicate a more balanced performance between sensitivity and specificity.

The highest G-Mean was obtained for CD, followed by HYP and STTC. MI resulted in the lowest G-Mean among the five categories, indicating a relatively greater difficulty in simultaneously maintaining sensitivity and specificity for this diagnostic class. Overall, the confusion matrix and G-Mean analysis provides a complementary view of the model behavior beyond ROC-AUC and highlights class-specific differences in threshold-based classification performance.

Table 5 Class-wise G-Mean values of the proposed HeartLang framework with ECG preprocessing for the representative seed-2 test run.

Diagnostic Class	G-Mean
NORM	0.7717
MI	0.6313
STTC	0.8173
CD	0.8542
HYP	0.8271

4.8 Effect of ECG Preprocessing

To quantify the contribution of the proposed ECG preprocessing pipeline, the final HeartLang Full Fine-Tuning model was evaluated with and without preprocessing while keeping the downstream training and evaluation protocol unchanged. The preprocessing pipeline consisted of band-pass filtering and lead-wise normalization before QRS tokenization.

The results show that preprocessing improved the overall discriminative performance of the HeartLang model. The Macro ROC-AUC

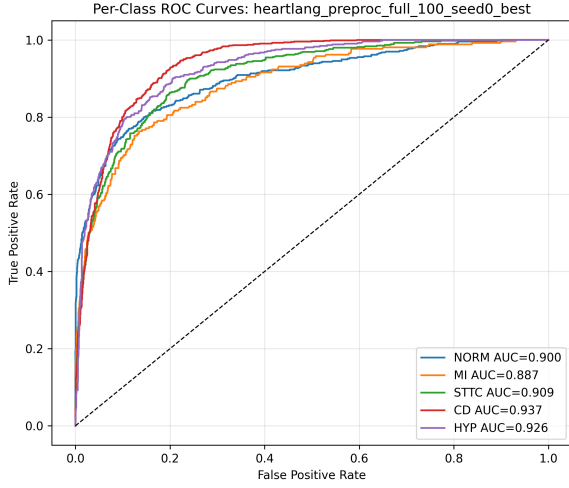
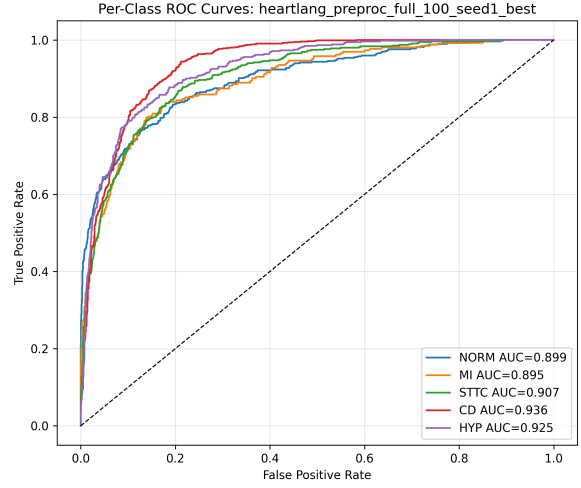
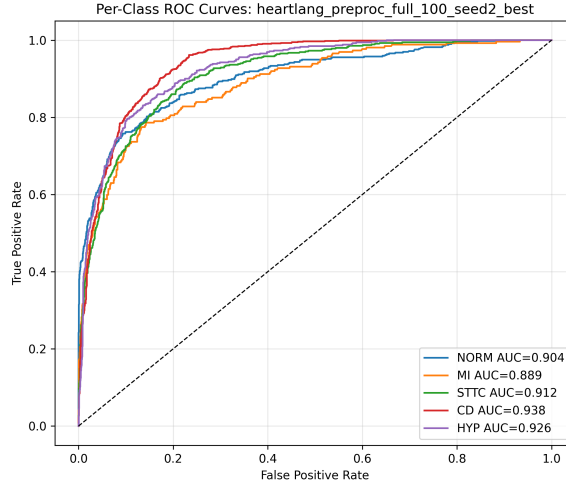
(a) Seed 0**(b) Seed 1****(c) Seed 2**

Fig. 6 Class-wise ROC curves for the preprocessed HeartLang Full Fine-Tuning model across three random seeds on the PTB-XL Superdiagnostic test set. Panels (a), (b), and (c) correspond to seeds 0, 1, and 2, respectively.

increased from 0.9020 ± 0.0013 without preprocessing to 0.9126 ± 0.0010 with preprocessing, corresponding to an absolute improvement of 0.0106 in ROC-AUC. Accuracy also increased from 0.5618 ± 0.0135 to 0.5779 ± 0.0093 , while Precision increased from 0.7358 ± 0.0172 to 0.7608 ± 0.0121 .

Figure 8 illustrates the ROC-AUC comparison.

The test loss decreased from 0.3174 ± 0.0121 to 0.3006 ± 0.0040 , indicating improved optimization and generalization under the preprocessed configuration. However, Recall decreased slightly from

0.6662 ± 0.0107 to 0.6549 ± 0.0099 , and the F1-score remained essentially unchanged, changing from 0.6792 ± 0.0051 to 0.6791 ± 0.0062 .

These results indicate that ECG preprocessing primarily improves the discriminative capability of the pretrained HeartLang representation, as reflected by the higher ROC-AUC, while its effect on threshold-dependent metrics is more mixed. The overall improvement in ROC-AUC suggests that reducing signal variability and standardizing the ECG inputs before QRS tokenization provides

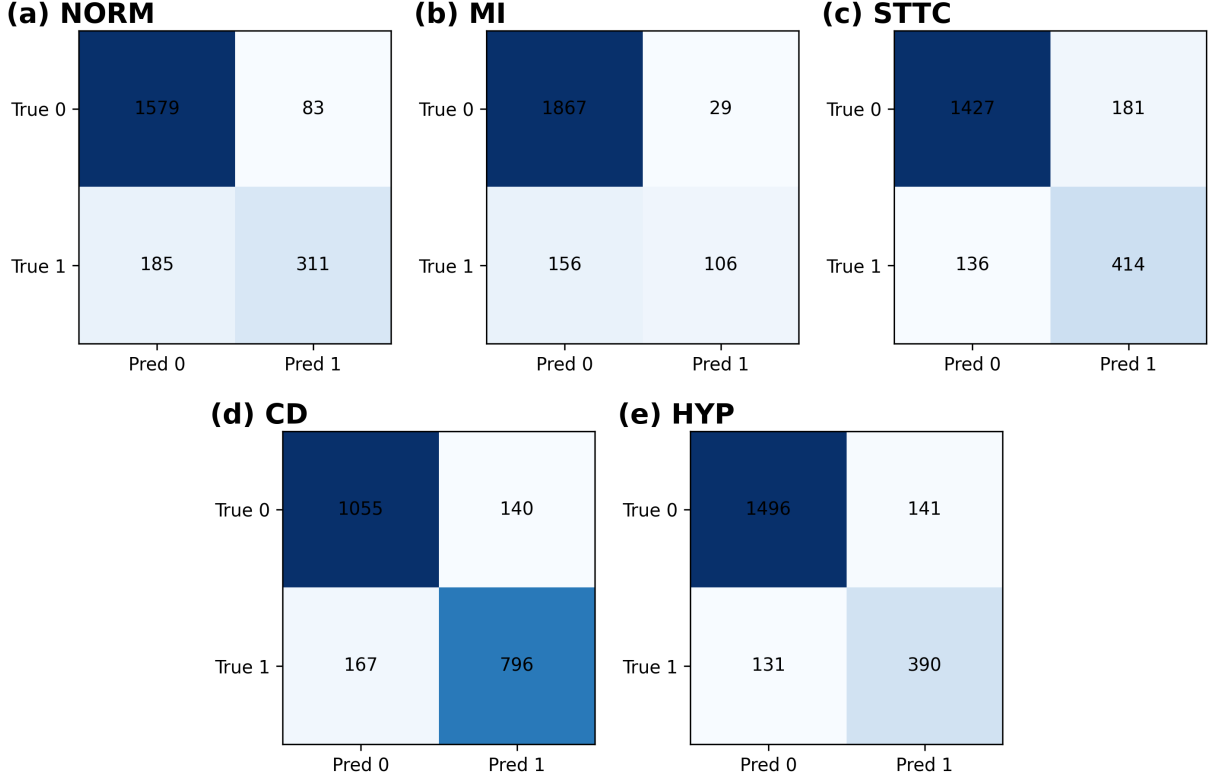


Fig. 7 Class-wise confusion matrices for the preprocessed HeartLang Full Fine-Tuning model on the PTB-XL Superdiagnostic test set. The matrices correspond to the representative seed-2 test run for NORM, MI, STTC, CD, and HYP.

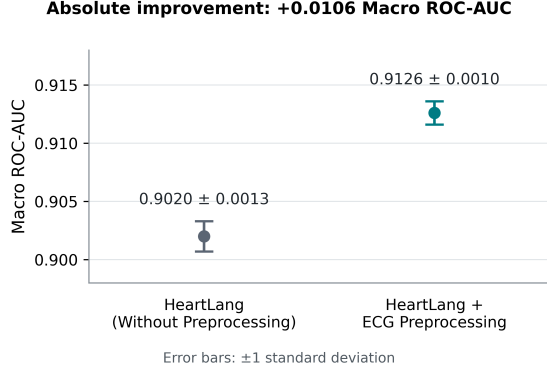


Fig. 8 Effect of the proposed ECG preprocessing pipeline on the Macro ROC-AUC of HeartLang Full Fine-Tuning.

a more suitable representation for downstream HeartLang fine-tuning.

Table 6 Effect of ECG preprocessing on HeartLang Full Fine-Tuning performance.

Metric	Without Preprocessing	With Preprocessing
ROC-AUC	0.9020 ± 0.0013	0.9126 ± 0.0010
Accuracy	0.5618 ± 0.0135	0.5779 ± 0.0093
Precision	0.7358 ± 0.0172	0.7608 ± 0.0121
Recall	0.6662 ± 0.0107	0.6549 ± 0.0099
F1-score	0.6792 ± 0.0051	0.6791 ± 0.0062
Loss	0.3174 ± 0.0121	0.3006 ± 0.0040

4.9 Discussion

The experimental results demonstrate that pre-trained ECG representations provided by HeartLang can be effectively adapted to the PTB-XL Superdiagnostic multi-label classification task. The proposed framework, which combines ECG preprocessing with Full Fine-Tuning, achieved a Macro ROC-AUC of 0.9126 ± 0.0010 on the held-out test set. The low variation across the three random seeds indicates stable performance under repeated experimental runs.

The fine-tuning ablation study showed that the extent of adaptation of the pretrained encoder has a substantial effect on downstream performance. Linear Probing achieved a Macro ROC-AUC of 0.8719 ± 0.0004 , demonstrating that the pretrained representations are already informative for the target task. However, Full Fine-Tuning increased the Macro ROC-AUC to 0.9020 ± 0.0013 and substantially improved the F1-score. In contrast, the Adapter Fine-Tuning configuration performed poorly in the evaluated setup and was therefore not continued for additional random seeds. These findings suggest that allowing the pretrained encoder to adapt fully to the target ECG distribution is more effective for this task than restricting encoder updates.

The baseline comparison further supports the effectiveness of the HeartLang-based approach. Among the conventional baseline architectures, ResNet achieved the strongest Macro ROC-AUC of 0.8868 ± 0.0009 , whereas the proposed preprocessed HeartLang framework reached 0.9126 ± 0.0010 . This improvement indicates that pretrained ECG-specific representations can provide a stronger basis for downstream classification than conventional architectures trained directly for the target task.

The data-efficiency analysis showed that the proposed framework maintained competitive performance when only a fraction of the available training data was used. Macro ROC-AUC increased from 0.8682 with 10% of the training data to 0.9126 with the complete training set. This behavior demonstrates that the pretrained model can extract useful information even under restricted labeled-data conditions, while additional labeled ECG data continues to improve downstream performance.

The class-wise analysis revealed consistently strong discrimination across all five Superdiagnostic categories. Conduction Disturbance (CD) achieved the highest ROC-AUC of 0.9369 ± 0.0009 , followed by Hypertrophy (HYP) with 0.9255 ± 0.0009 . Myocardial Infarction (MI) produced the lowest class-wise ROC-AUC at 0.8908 ± 0.0040 , indicating that this category remains comparatively more challenging for the model. The confusion matrix and G-Mean analysis further showed differences in the balance between sensitivity and specificity across diagnostic categories.

The comparison of preprocessing configurations showed that signal preprocessing improved the Macro ROC-AUC from 0.9020 ± 0.0013 to 0.9126 ± 0.0010 , corresponding to an absolute improvement of 0.0106. Accuracy, Precision, and test loss also improved, while Recall and F1-score remained approximately unchanged. This indicates that preprocessing mainly enhanced the discriminative quality of the representations rather than uniformly improving every threshold-dependent metric.

Overall, the results suggest that the combination of pretrained ECG representation learning, full downstream adaptation, and appropriate ECG signal preprocessing provides an effective framework for multi-label ECG classification. At the same time, the variation observed across diagnostic categories and the remaining gap between some threshold-dependent metrics highlight the need for further investigation of class-specific behavior and more extensive evaluation on additional datasets.

5 Conclusion

This study presented a HeartLang-based framework for multi-label ECG classification on the PTB-XL Superdiagnostic dataset. The framework combines a pretrained HeartLang representation with ECG signal preprocessing and Full Fine-Tuning for downstream diagnostic classification across five Superdiagnostic categories.

The experimental results demonstrated that the proposed preprocessed HeartLang framework achieved a Macro ROC-AUC of 0.9126 ± 0.0010 on the held-out test set. Compared with the corresponding HeartLang configuration without preprocessing, preprocessing improved the Macro ROC-AUC by 0.0106. The ablation study further showed that Full Fine-Tuning provided stronger downstream performance than Linear Probing and the evaluated Adapter Fine-Tuning configuration, highlighting the importance of adapting the pretrained encoder to the target task.

The baseline comparison demonstrated that the proposed framework achieved higher Macro ROC-AUC than the evaluated conventional architectures, including Linear Classifier, MLP, CNN, GRU, and ResNet. The data-efficiency analysis also showed that HeartLang maintained competitive performance with reduced amounts of

labeled training data, while increasing the available training data continued to improve performance. Class-wise analysis demonstrated strong discrimination across all five diagnostic categories, with some variation in performance between individual classes.

Overall, the findings indicate that pretrained ECG representation learning, combined with appropriate signal preprocessing and downstream Full Fine-Tuning, provides an effective approach for multi-label ECG classification. The study also provides an empirical basis for further investigation of preprocessing strategies, efficient adaptation methods, and data-efficient training of pretrained ECG models on diverse clinical datasets.

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